

EQUESTRIAN WASTELAND SIM — MASTER INDEX (EWS-INDEX/1)

Stacks on GHOST.md (Build 11) · Universal AI Architecture & Chunk-Scanning Registry

0. ARCHITECTURAL HANDSHAKE & STACKING RULE

- **Layer 0 (Core Safety & Protocol):** `GHOST.md` — R1–R7 rules, Health line, session logs, touch-first constraints (≥ 44 px iPad).
- **Layer 1 (System & Chunk Index):** `MASTER_INDEX.md` (This file) — Unifies the entire project architecture, 3D character mesh transition, shell layers, and modular chunk-scanning index for AI models.
- **Layer 2 (Roadmap & Deep Lore):** `PROJECT_BIBLE.md` — Packages A through O.
- **Layer 3 (Tool & Studio Rigs):** `Stable-Tec_Master_DevKit_Studio.html` — Live 3D DevKit Studio with anatomical rigs, fluid mechanics, and transform tools.

1. CHUNK-SCANNING REGISTRY FOR AI SESSIONS

To prevent context window overflow, browser freezes, and "Failed to Load" crashes in AI Studio / Claude / ChatGPT, all architecture modules are organized into distinct scan chunks. An AI can read or query specific chunk tags without loading unnecessary files.

Chunk Tag	Topic / Domain	Primary File / Target	Scope & Summary
[CHUNK:3D_MESH_RIG]	3D Character Mesh & Rigging	Stable-Tec_Master_DevKit_Studio.html	Transition from 2.5D implicit bean hulls to full 3D polygon mesh characters operating within the 2.5D isometric world.
[CHUNK:SHELL_LAYERS]	Unified Shell Layering	Stable-Tec_Master_DevKit_Studio.html	Base Avatar, Organs, Armor, Clothes, Accessories. Layer isolation, see-through

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			ghost shells, and layer editing.
[CHUNK:SPHERE_RE SPECT]	Surface Constraint Engine	Stable-Tec_Master_DevKit_Studio.html	Glide Mode (Wrap Sticker) vs Deform Mode (Mesh Sculpt) for surface-attached exit nodes.
[CHUNK:ANATOMY_F LUIDS]	Internal Viscera & Excretion	Stable-Tec_Master_DevKit_Studio.html	Esophagus inlet, Stomach bag, Bowel exit, Bladder, Uterus/Repro canal, Male ventral sheath exit, bag-in-a-bag limiter.
[CHUNK:CONSUMABLE E_TAX]	Ingestion & Contraband	PROJECT_BIBLE.md (Package O)	Nutrient Paste slurry, liquid vs solid taxonomy, swallowed lockpick jailbreak pipeline.
[CHUNK:WORLD_ISO PIPE]	2.5D Isometric World Pipeline	src/sim/simEngine.ts, iso_asset_viewer.html	64x32 tile coordinates, 1 block = 32px height, 3D character mesh projection into 2.5D scene.
[CHUNK:CRASH_RED UNDANCY]	Session & Cloud Redundancy	GHOST.md, Google Drive Master Vault	Auto-syncing state, cache key bumping (st_block7), error trapping, zero data loss.
[CHUNK:ZERO_BAR_UI]	Master UI Law (Zero-Bar)	EWS_with_Build.html, EWS_MASTER_AI_HANDOFF.md	100% borderless 3D viewport, 3 corner glass orbs, draggable Revolver 6-shooter dial, sliding sheets, Nude Base View.

Chunk Tag	Topic / Domain	Primary File / Target	Scope & Summary
[CHUNK:WILD_PONY_LORE]	Equine Sociology & Childcare	EWS_MASTER_AI_HA NDOFF.md, PROJECT_BIBLE.md	Settler vs Wild/Feral waste ponies (hygiene, outdoor elimination), foal diapers & potty training progression (0-2).
[CHUNK:TOON_HAIR]	Two-Piece Mane & Shape Law	EWS_with_Build.h tml, EWS_MASTER_AI_HA NDOFF.md	Anti-photorealism law, chunky volumetric toon ribbons, Nape/Crest neck mane + Crown/Forelock brow mane + S-curve tail.

2. [CHUNK:3D_MESH_RIG] 3D CHARACTER MESH IN 2.5D WORLD

- **Core Paradigm Shift:** The character rig is transitioning from 2.5D sphere hulls to true 3D mesh geometry (vertices, edges, faces, polygon contours) with bone skinning.
 - **Perspective Integration:** Characters render in true 3D coordinates \$(x, y, z)\$ projected into the engine's fixed isometric 2.5D grid (\$64 \times 32\$ pixel diamond tiles; \$z\$-axis represents 32px vertical blocks).
 - **Sub-Object Editing Modes:**
 - 🦴 **Bone:** Select and manipulate joints/bones and parent-child hierarchies.
 - ✦ **Vertex:** Select and nudge vertex clusters.
 - – **Edge:** Edge loop weighting and contour creasing.
 - ◻ **Face:** Surface face picking and decal/sticker wrapping.
 - **Outline & Aesthetic Rule:** Tube-cute flat vector finish. Outlines only generate on the union outer silhouette and sharp silhouette breaks; internal shared faces blend cleanly.
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3. [CHUNK:SHELL_LAYERS] UNIFIED SHELL LAYERS

All character elements and garments share the underlying 3D skeletal mesh:

1. 🎨 **Base Avatar**: Core pony body mesh (head, muzzle, chest, rear, limbs, ears, wings).
2. 🦴 **Organs**: Internal visceral anatomy rig, fluid containers, and hollow conduits.
3. 🛡️ **Armor**: Barding, saddles, scrap metal plates, helmets (deforms with parent bones).
4. 👕 **Clothes**: Undergarments, flight suits, lab coats, vault jumpsuits.
5. 🎒 **Accessories**: Saddlebags, holsters, goggles, harnesses, tools.

Layer Isolation & See-Through Ghosting

- When a shell layer is isolated (e.g. 🦴 **View: Organs Only**), non-active layers are hidden to eliminate visual obstruction during modeling.
 - **Surgical Reference Ghost**: When the outer body is hidden, a partial transparent see-through ghost shell of the **rear** sphere (the butt) and **chest** renders with cyan dashed contour lines (#38bdf8) and 3D depth equator/meridian rings.
 - **Exit Aperture Collars**: Glowing surface rings mark the exact spot where internal tubes breach the outer surface mesh.
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4. [CHUNK:SPHERE_RESPECT] SURFACE CONSTRAINT ENGINE

Allows precise placement and sizing of surface-attached nodes (exit sphincters, inlets, decal anchors):

- 🌐 **Glide Mode (Respect Sphere / Wrap Sticker)**:
 - Movement is mathematically projected onto the parent sphere's surface ($\$P.r \times P.sq\$$).
 - Moving Move Gizmo handles ($\$X, Y, Z\$$) or direct touch dragging glides the node across latitude and longitude along the curved outer face, maintaining constant distance from center.
 - The parent sphere geometry is strictly preserved and not deformed.
 - 🦋 **Deform Mode (Mesh Sculpt)**:
 - Moving the node outward stretches parent squash ($\$sq\$$) and pulls the mesh out into an anatomical cone or nozzle. Pushing inward creates an indentation.
 - **Aperture Sizing**: Scaling the node adjusts the opening diameter ($\$r\$$) from tightly closed ($\$0.02\$$) to fully dilated ($\$0.30\$$).
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5. [CHUNK:ANATOMY_FLUIDS] INTERNAL VISCERA & DUAL EXCRETION

- **1 Front Inlet:** Hollow 3-stage esophagus tube (`esoph_top` $\rightarrow$ `esoph_mid` $\rightarrow$ `esoph_low`) from mouth/head into stomach.
 - **Stomach Bag:** Centered in visceral abdominal space; dynamic volume (0 to $60 \times \{ \text{units} \}$).
 - **Exit 1 (Solid Digestive Waste):** Stomach $\rightarrow$ `bowel_mid` $\rightarrow$ `bowel_exit` (welded to posterior outer curvature of rear sphere).
 - **Exit 2 (Fluid Renal Waste):** Pelvic floor `bladder_bag` $\rightarrow$ `urethra_exit` (Mare) or forward through `sheath_conduit` $\rightarrow$ `male_ventral_exit` on lower belly (Stallion).
 - **Exit 3 (Reproductive Birth Canal - Mare):** Bicornuate `uterus_bag` with dual uterine horns (`horn_L/R`) and ovaries (`ovary_L/R`) $\rightarrow$ `repro_exit` out back.
 - **Male Reproductive Kit (Stallion):** External scrotal testes (`testis_L/R`) and spermatic duct (`vas_deferens`) connecting to ventral exit.
 - **Bag-in-a-Bag Peritoneal Limiter:** The stomach, bladder, and uterus expand inside a fixed peritoneal boundary ($100 \times \{ \text{units} \}$). When combined volume exceeds capacity, pressure forces external belly deformation.
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6. [CHUNK:CONSUMABLE_TAX] INGESTION, PASTE & CONTRABAND

- **Nutrient Paste Slurry:** Liquid-state food flagged as fluid consumable that satisfies hunger without requiring solid mastication, allowing paste-dependent settlers to survive.
 - **Ingested Contraband Pipeline:** Non-digestible items (e.g. lockpicks, brass keys, data chips) swallowed prior to capture pass through the digestive tract and exit via `bowel_exit` during elimination to pick cell locks.
 - **Reverse Peristalsis:** Emergency stomach purge ejects non-digested items and poisons via mouth.
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7. [CHUNK:CRASH_REDUNDANCY] REDUNDANCIES & STATE ARCHIVAL

- **Single Source of Truth:** `GHOST.md` + Google Drive Master Vault.

- **Cache Invalidation:** LocalStorage cache keys are bumped (`st_block7`) on major architectural changes to prevent stale browser state.
- **Removable DevKit Rule:** DevKit tools run self-contained in single HTML files, fully operational on iPad (Safari/Textastic) and desktop browsers with offline capabilities.

9. [CHUNK:ZERO_BAR_UI] MASTER UI SPECIFICATION: THE ZERO-BAR LAYOUT LAW

- **Core Law:** Exactly 0 persistent top or bottom chrome bars. 100% borderless 3D viewport.
- **Outer Margin (--m = 12px):** Minimal corner glass orbs (>=44px touch targets):
 - Bottom-Left (`#mode`): Pause / Live toggle.
 - Bottom-Right (`#time`): Simulation speed (1x, 2x, 5x).
 - Top-Right (`#sub`): Cinematic low-angle ground camera toggle.
 - Left-Side (`#rag`): Active Ragdoll physics toggle.
 - Left-Side (`#bld`): Stage 3 Build Mode iframe overlay toggle.
 - Top-Left (`#layer-dropdown`): Hierarchical Layer Isolation dropdown.
- **Inner Margin (--mi = 60px):**
 - **Draggable Revolver 6-Shooter Dial:** 150x150px circular touch hub with central clock/state and 6 radial bullet buttons. Context auto-maps between LIVE, BUILD, and SELECT modes. Snaps within safe margins.
 - **Sliding Glass Sheets (#sheet):** Slide up from bottom on-demand when summoned. Auto-dismiss on click-off into empty ground.
- **Layer Isolation:** Layers 1–5 (Deep Viscera, Skeletal Rig, Skin, Clothing, Armor) plus Nude Base View (clean chibi form, organs hidden inside, no skeleton lines).
- **Action & Behavioral Director:** Initiator -> Target pairing (Me, Charlie, Cindy, Mirror, Self), action palette (Hug, Kiss, Salute, Groom, Brawl), physical intensity (25% to 100%), and obsession loops (Once, Doting, Obsessive Loop).

10. [CHUNK:WILD_PONY_LORE] EQUINE SOCIOLOGY & CHILDCARE MECHANICS

- **Settler & Vault Ponies:** Civilized, organized, value indoor plumbing, waste containment, and clean living environments.
- **Wild & Feral Wasteland Ponies:** Un-housebroken, eliminate freely outdoors in dirt, mud, and water ("wild ponies are dirty and pee"). Require socialization or taming.
- **Childcare & Foal Mechanics:** Diapers equipment for young foals. Progressive potty-training skill levels: Level 0 (Untrained/needs diapers), Level 1 (Rudimentary), Level 2 (Housebroken). Sanitation penalties if soiled.

11. [CHUNK:TOON_HAIR] STYLIZED TOON SHAPE LANGUAGE & TWO-PIECE MANE LAW

- **Anti-Photorealism Rule:** Strictly avoid raw photographic textures, realistic skin pores, photorealistic PBR noise, or complex micro-strand hair (NO 400 loose hair strands!).
- **Shape Language:** Clean, bold, charming cartoon shapes with soft toon lighting and harmonizing gradients.
- **Two-Piece Modular Mane Architecture:**
 1. Nape / Crest Mane: Anchors along dorsal neck ridge, behind skull, over withers (Rainbow Dash flowing locks or Scootaloo tomboy spikes).
 2. Crown / Forelock Mane: Forehead/brow fringe framing the face and eyes.
 3. S-Curve Volumetric Tail: Anchored at sacral dock on pelvis.
- **Volumetric Toon Ribbons:** Modeled as chunky, tapered spline ribbons with thick roots and soft rounded tips.